

Null Invariance: An Important Property

- Why is null invariance crucial for the analysis of massive transaction data?
 - Many transactions may contain neither milk nor coffee!

milk vs. coffee contingency table

	<i>milk</i>	$\neg milk$	Σ_{row}
<i>coffee</i>	<i>mc</i>	$\neg mc$	<i>c</i>
$\neg coffee$	<i>m</i> $\neg c$	$\neg m$ $\neg c$	$\neg c$
Σ_{col}	<i>m</i>	$\neg m$	Σ

- Lift and χ^2 are not null-invariant: not good to evaluate data that contain too many or too few null transactions!
- Many measures are not null-invariant!

Null-transactions
w.r.t. *m* and *c*

Data set	<i>mc</i>	$\neg mc$	<i>m</i> $\neg c$	$\neg m$ $\neg c$	χ^2	<i>Lift</i>
<i>D</i> ₁	10,000	1,000	1,000	100,000	90557	9.26
<i>D</i> ₂	10,000	1,000	1,000	100	0	1
<i>D</i> ₃	100	1,000	1,000	100,000	670	8.44
<i>D</i> ₄	1,000	1,000	1,000	100,000	24740	25.75
<i>D</i> ₅	1,000	100	10,000	100,000	8173	9.18
<i>D</i> ₆	1,000	10	100,000	100,000	965	1.97

Comparison of Null-Invariant Measures

- ❑ Not all null-invariant measures are created equal
- ❑ Which one is better?
 - ❑ $D_4 - D_6$ differentiate the null-invariant measures
 - ❑ Kulc (Kulczynski 1927) holds firm and is in balance of both directional implications

2-variable contingency table

	<i>milk</i>	$\neg milk$	Σ_{row}
<i>coffee</i>	<i>mc</i>	$\neg mc$	<i>c</i>
$\neg coffee$	<i>m</i> $\neg c$	$\neg m \neg c$	$\neg c$
Σ_{col}	<i>m</i>	$\neg m$	Σ

All 5 are null-invariant

Data set	<i>mc</i>	$\neg mc$	<i>m</i> $\neg c$	$\neg m \neg c$	<i>AllConf</i>	Jaccard	<i>Cosine</i>	<i>Kulc</i>	<i>MaxConf</i>
D_1	10,000	1,000	1,000	100,000	0.91	0.83	0.91	0.91	0.91
D_2	10,000	1,000	1,000	100	0.91	0.83	0.91	0.91	0.91
D_3	100	1,000	1,000	100,000	0.09	0.05	0.09	0.09	0.09
D_4	1,000	1,000	1,000	100,000	0.5	0.33	0.5	0.5	0.5
D_5	1,000	100	10,000	100,000	0.09	0.09	0.29	0.5	0.91
D_6	1,000	10	100,000	100,000	0.01	0.01	0.10	0.5	0.99

Subtle: They disagree on those cases